

TALAMANDA AI

TRUST LAYER PLATFORM

Technical Architecture & Flow Documentation

Aegis Agentic Swarm • The Librarian • The Adversary • The Notary
Independent AI Governance | NIST AI RMF | ISO 27001 | Sovereign Deployment

CONFIDENTIAL

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Talamanda AI

Internal Use Only

1. Executive Summary

Talamanda AI is an independent, behavior-level trust and governance platform for enterprise AI deployments. Unlike traditional security tools that audit infrastructure, Talamanda's Aegis Swarm operates as a real-time referee - continuously monitoring, red-teaming, and certifying the safety and compliance posture of AI agents deployed in high-stakes business environments.

Core Value Proposition
→ Independent audit layer completely decoupled from model providers (Big Tech cannot audit their own models)
→ Behavior-level AI governance - not just infrastructure scanning
→ Sovereign, on-premise deployment - sensitive metadata never leaves organizational boundaries
→ Real-time compliance mapping to NIST AI RMF, ISO 27001, and EU AI Act
→ Cryptographically signed audit evidence accepted by regulators and customers

The platform is purpose-built for industries where AI agent failures carry direct financial, regulatory, and operational consequences: Retail, Manufacturing, Semiconductors, Logistics, and Banking.

2. Architecture Overview

2.1 The Trust Layer Paradigm

Talamanda inserts a sovereign, stateful middleware layer - the Trust Layer - between an organization's AI models and its live business operations. This is architecturally distinct from API gateways or network proxies; it is a full agentic control plane that observes, challenges, and certifies AI behavior.



2.2 High-Level Component Layers

Layer	Component	Responsibility
Ingestion Layer	Regulation Feed Ingester	Continuously pulls AI law, EU AI Act, NIST updates
Ingestion Layer	Control Mapper	Maps regulatory requirements to technical controls
Adversarial Layer	Prompt Injection Engine	24/7 automated red-teaming of deployed agents
Adversarial Layer	Logic Manipulation Prober	Tests semantic reasoning integrity under manipulation
Evidence Layer	Evidence Collector	Aggregates behavioral telemetry and test outcomes
Evidence Layer	Cryptographic Signer	Signs compliance artifacts for legal admissibility
Reporting Layer	Compliance Dashboard	Real-time NIST/ISO posture visualization
Reporting Layer	Report Generator	Generates regulator-ready audit packages
Deployment Layer	Trust Node Manager	Manages on-premise / private cloud sovereign deployments

3. The Aegis Agentic Swarm

The Aegis Swarm is the operational core of the Talamanda platform. It is composed of three specialized autonomous agents that operate concurrently within the customer's sovereign deployment boundary. Each agent has a distinct responsibility, and together they form a closed-loop AI governance system.

 LIBRARIAN	Regulatory Intelligence Agent - Continuously ingests global AI regulations: AI Basic Law, EU AI Act, NIST AI RMF, ISO 27001, SOC 2 - Parses and versions regulatory change sets, tagging changes by industry vertical and risk category - Maintains a live control library mapping each regulation clause to a specific, testable technical control - Exposes the control library as a query API consumed by both the Adversary and the Notary agents - Triggers re-assessment events when regulatory updates alter existing control mappings
 ADVERSARY	Red-Team & Vulnerability Agent - Operates as a 24/7 automated adversarial tester against deployed AI agents in a sandboxed evaluation harness - Executes prompt injection attacks: direct injection, indirect injection, jailbreak sequences, role-play bypasses - Performs logic manipulation probes: goal mis-specification, false premise injection, constraint overriding - Runs supply-chain-specific attack scenarios: illegal routing shortcuts in logistics, bypassing safety protocols in manufacturing - Each test case is pulled from the Librarian's control library, ensuring adversarial coverage maps to regulatory obligations - Vulnerability findings are streamed in real time to the Notary for evidence packaging
 NOTARY	Compliance Evidence & Certification Agent - Aggregates behavioral telemetry, test results, and control mapping outputs from the Librarian and Adversary - Produces cryptographically signed compliance attestation reports (SHA-256 + RSA-2048 evidence chains) - Generates regulator-ready audit packages aligned to NIST AI RMF tiers and ISO 27001 Annex A controls - Maintains an immutable evidence ledger with full chain of custody for each compliance assertion - Exposes a Trust Score API enabling real-time safety certification queries for Board, Legal, and Risk stakeholders - Issues the 'Safety Certificate' - the human-readable governance artifact required before AI agent go-live

4. Technical Data Flow

4.1 Primary Trust Evaluation Pipeline

The following describes the end-to-end data flow from AI agent action to compliance certification:

1. AI Agent Emits Action - An enterprise AI agent (e.g., a trade finance decision agent or logistics routing agent) initiates an action or produces an output within the business system.
2. Trust Layer Intercept - The Talamanda Trust Node intercepts the agent's action payload before it reaches production systems. The payload is serialized and timestamped.
3. Librarian Control Lookup - The payload's context (industry vertical, agent role, data type) is matched against the Librarian's live control library. Applicable regulatory controls are identified.
4. Adversary Evaluation - The Adversary agent executes the relevant adversarial test suite against the intercepted payload in a sandboxed harness, checking for prompt injection vulnerabilities and logic integrity violations.
5. Risk Scoring - A composite risk score is calculated from: (a) control coverage delta, (b) adversarial test pass rate, (c) behavioral drift from baseline, and (d) historical violation frequency.
6. Pass / Hold Gate - If risk score is below threshold, the action is released to production. If above threshold, the action is held, flagged, and routed to human-in-the-loop review.
7. Notary Evidence Capture - Regardless of gate outcome, the Notary captures the full evaluation record, signs it cryptographically, and appends it to the immutable evidence ledger.
8. Dashboard & Alert Update - The compliance dashboard is updated in real time. Threshold breaches trigger alerts to configured stakeholder channels.

4.2 Regulatory Update Flow

Librarian Regulation Ingestion Pipeline
1. External regulation feeds (EU AI Act updates, NIST AI RMF amendments, national AI laws) are polled on a configurable schedule (default: every 6 hours).
2. Change detection engine diffs new content against the current control library version.
3. Changed clauses are parsed, tagged by: jurisdiction industry vertical risk tier affected control IDs.
4. Updated mappings are versioned and published to the internal Control Library API (v-stamped, rollback-capable).
5. Re-assessment triggers are emitted to the Adversary for any controls whose test coverage must be updated.
6. Compliance dashboard reflects updated posture within the next evaluation cycle.

4.3 Adversarial Red-Team Execution Flow

Adversary Agent Test Execution Pipeline
1. Adversary pulls active test cases from the Librarian control library, filtered by deployment context.
2. Test harness initializes an isolated evaluation environment mirroring the production AI agent's interface.
3. Prompt injection battery executes: direct, indirect, multi-turn, and role-play injection vectors.
4. Logic manipulation battery executes: false premise injection, goal drift probes, constraint override attempts.
5. Supply-chain-specific probes execute based on vertical configuration (e.g., routing safety for logistics, production line bypass for manufacturing).
6. Each test result is scored: PASS FAIL PARTIAL - with full attack payload and model response captured.
7. Findings streamed to Notary in real time via internal event bus (topic: adversary.findings).

5. Sovereign Deployment Architecture

5.1 Trust Node Topology

Talamanda operates on a Trust Node model. Each Trust Node is a self-contained, sovereign deployment unit that runs all three Aegis agents within the customer's own infrastructure boundary. No sensitive metadata, AI payloads, or business telemetry transits Talamanda's infrastructure.

Deployment Mode	Infrastructure	Data Residency
On-Premise	Customer-managed bare metal or VM	100% within customer datacenter
Private Cloud	Customer VPC (AWS, Azure, GCP)	Isolated VPC, no cross-tenant routing
Air-Gapped	No external network connectivity	Fully isolated, manual update packages
Hybrid	On-premise data + cloud compute	Data at rest on-premise, processing isolated

5.2 Network Architecture

- All inter-agent communication uses mTLS (mutual TLS 1.3) within the Trust Node boundary.
- The Control Library API is exposed only on the node's internal network segment - never externally reachable.
- Regulation feed ingestion is the only outbound connection: HTTPS to configured regulatory data sources, with optional egress proxy.
- The compliance dashboard UI is served over HTTPS within the customer's network; no data leaves via the UI layer.
- Cryptographic signing keys are generated and stored within the Trust Node using HSM-backed key storage (or software HSM in cloud deployments).

5.3 Data Classification & Handling

Data Type	Classification	Handling
AI agent input/output payloads	CONFIDENTIAL	Processed in-node, never exfiltrated, purged per retention policy
Adversarial test cases	INTERNAL	Generated from control library, stored in-node
Compliance evidence records	LEGAL-HOLD	Immutable ledger, signed, retained per regulatory requirement
Regulation feeds (inbound)	PUBLIC	Pulled from external sources, versioned locally
Trust Score / Audit Reports	CONFIDENTIAL	Encrypted at rest, shared only with authorized stakeholders

6. Compliance Framework Mapping

6.1 NIST AI Risk Management Framework (AI RMF)

The Aegis Swarm maps directly to all four NIST AI RMF core functions:

NIST AI RMF Function	Aegis Agent	Implementation
GOVERN	Librarian + Notary	Policy library maintained; governance evidence generated per deployment
MAP	Librarian	Regulations mapped to controls; controls mapped to agent behaviors
MEASURE	Adversary + Notary	Continuous behavioral testing; quantified risk scores and findings
MANAGE	Adversary + Notary	Automated remediation triggers; hold gates; signed incident records

6.2 ISO 27001 Annex A Automation

The Notary agent automates evidence collection for the following ISO 27001 Annex A control domains relevant to AI deployments:

- A.5 - Organizational Controls: AI governance policy documentation, roles and responsibilities
- A.8 - Technological Controls: AI system access control, software integrity, vulnerability management
- A.12 - Operations Security: Monitoring, logging, and audit log protection for AI agent operations
- A.16 - Information Security Incident Management: Adversary-detected AI failure incident records
- A.18 - Compliance: Regulatory obligation mapping and evidence of compliance assertion

6.3 EU AI Act Readiness

For customers operating in EU jurisdictions or serving EU customers, the Librarian maintains a dedicated EU AI Act control set covering:

- High-risk AI system classification requirements (Annex III)
- Technical documentation obligations (Article 11)
- Transparency and explainability requirements (Article 13)
- Human oversight provisions (Article 14)
- Accuracy, robustness, and cybersecurity obligations (Article 15)

7. Unique Selling Proposition & Differentiation

7.1 The Structural USP: Independence

The single most defensible competitive moat for Talamanda is structural independence. Big Tech AI providers (OpenAI, Google, Anthropic, Microsoft) are constitutionally incapable of credibly auditing the safety of their own models. This creates a permanent, governance-mandated market for an independent referee - a role Talamanda is purpose-built to occupy.

Why Independence Cannot Be Replicated by Incumbents
→ Model providers have a commercial incentive to minimize reported vulnerabilities in their own systems.
→ Regulatory bodies globally are moving toward mandatory third-party AI audits (EU AI Act Article 43, emerging AI Basic Law provisions).
→ Board-level AI governance requires separation of duties: the entity that builds the AI cannot be the entity that certifies it.
→ Talamanda's sovereign deployment model means the auditor never touches the production model - it audits behavior outputs only.

7.2 Competitive Landscape Comparison

Capability	Traditional Security Tools	Talamanda Aegis Swarm
Infrastructure audit	✓ Yes	✓ Yes (baseline)
AI behavior audit	X No	✓ Core capability
Real-time red-teaming	X No	✓ 24/7 Adversary agent
Regulatory auto-mapping	X No	✓ Librarian agent
Cryptographic evidence	Partial (logs only)	✓ Full chain of custody
Sovereign deployment	X Typically SaaS	✓ On-premise / private cloud
Model-provider independence	N/A	✓ Structural separation
Supply chain attack simulation	X No	✓ Vertical-specific probes
Board-ready Safety Certificate	X No	✓ Notary-issued, signed

7.3 The 'Referee' Brand Position

Talamanda's market positioning as the AI Referee is strategically sound for the following reasons:

- Neutrality is monetizable: regulators, insurers, and enterprise legal teams will pay a premium for an auditor with no stake in the model's commercial success.
- The referee analogy is universally understood by Board and C-suite audiences who do not have deep AI technical knowledge.
- As AI liability law matures, the Safety Certificate issued by an independent referee will carry legal evidentiary weight - creating switching costs and long-term lock-in.
- The referee position is defensible against Big Tech expansion: their market presence in the model layer is precisely what disqualifies them from the auditor role.

8. Security Model

8.1 Threat Model Scope

The Talamanda Trust Layer is itself a security-critical system and is designed with a zero-trust internal architecture. The following threat categories are in scope for the platform's own security model:

- Compromise of the Trust Node by a malicious insider or external attacker attempting to tamper with audit results.
- Poisoning of the Librarian's regulation feed to suppress coverage of specific attack vectors.
- Adversary agent evasion - AI models trained to pass Adversary tests while remaining vulnerable in production.
- Evidence tampering - modification of the Notary's immutable ledger post-signing.
- Side-channel exfiltration of sensitive AI payloads via the Trust Node's network interface.

8.2 Security Controls

Threat	Control
Trust Node tampering	Signed software delivery, integrity attestation on startup (TPM/Secure Boot)
Regulation feed poisoning	Feed source pinning, cryptographic signature verification on feed updates
Adversary evasion	Adversary test cases are versioned, randomized, and rotated; behavioral drift detection supplements explicit test pass/fail
Evidence tampering	Append-only evidence ledger with cryptographic hash chaining; Notary signing keys in HSM
Payload exfiltration	Egress filtering at Trust Node network boundary; no outbound connections except regulation feed pulls

9. Metrics, Outcomes & KPIs

9.1 Platform Performance Metrics

Metric	Target	Description
Compliance overhead reduction	90%	Manual audit effort eliminated via automation
Red-team evaluation latency	< 200ms p99	Time from agent action intercept to gate decision
Regulation feed refresh lag	< 6 hours	Lag between regulatory publication and control library update
Evidence ledger integrity	100%	Hash chain verification pass rate
Adversary test coverage	> 95% of controls	Fraction of Librarian controls with active adversarial test cases
Trust Score availability	99.9% uptime	API availability for real-time safety queries

9.2 Customer Outcomes

- Board Confidence: A signed Safety Certificate from an independent referee satisfies Board-level AI governance requirements before go-live.
- Legal Risk Reduction: Cryptographically signed evidence records provide a defensible compliance posture in regulatory investigations and litigation.
- Global Market Access: NIST AI RMF and ISO 27001 alignment enables enterprise sales to US Federal, EU, and enterprise customers with third-party AI audit requirements.
- Operational Velocity: 90% reduction in manual compliance overhead allows AI teams to ship faster without accumulating regulatory debt.
- Supply Chain Resilience: Adversary probes specific to logistics and manufacturing prevent AI-driven operational failures and safety incidents.

10. Implementation Roadmap Alignment

10.1 Current State (audit-swarm v1.0)

The audit-swarm repository represents the foundational implementation of the Aegis Swarm architecture. The current build establishes:

- Core agent scaffolding for the Librarian, Adversary, and Notary roles
- Initial control library schema and regulatory ingestion pipeline
- Prompt injection and logic manipulation test harness (Adversary core)
- Cryptographic signing infrastructure for the Notary
- Trust Node deployment configuration for on-premise environments

10.2 Path to Production Readiness

Phase	Milestone	Deliverable
Phase 1 - Core	Adversary v1: prompt injection battery complete	20+ injection vectors tested, scored, logged
Phase 1 - Core	Librarian v1: NIST AI RMF control library seeded	Full RMF function-to-control mapping live
Phase 2 - Evidence	Notary v1: cryptographic evidence chain	SHA-256 hash chain + RSA signing operational
Phase 2 - Evidence	Dashboard v1: real-time posture view	NIST/ISO coverage visualization
Phase 3 - Sovereign	Trust Node packaging: Docker / on-prem	One-command sovereign deployment
Phase 3 - Sovereign	EU AI Act control library expansion	High-risk system controls coverage
Phase 4 - Enterprise	Safety Certificate issuance workflow	Board-ready PDF + API endpoint
Phase 4 - Enterprise	Vertical-specific Adversary packs	Logistics, manufacturing, banking probe sets

Development Philosophy
The audit-swarm implementation follows an agent-first, evidence-native design philosophy.
Every component is built to be auditable itself - the Trust Layer trusts nothing, including its own subcomponents.
Sovereign deployment is a first-class constraint, not an afterthought: no feature is designed that requires data to leave the Trust Node.